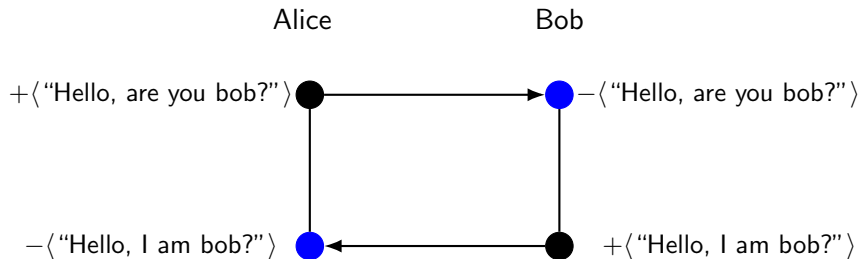


Session 3: CPSA Etymology and Attacks in CPSA

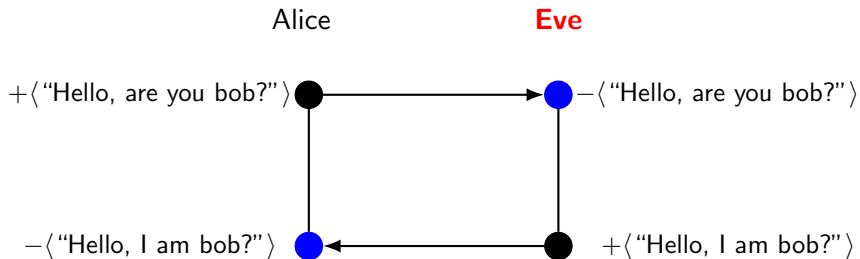
Slides prepared by UMBC Protocol Analysis Lab

Last edited June 26, 2026

A completed protocol?



Strands can lie



CPSA adversarial strand actions

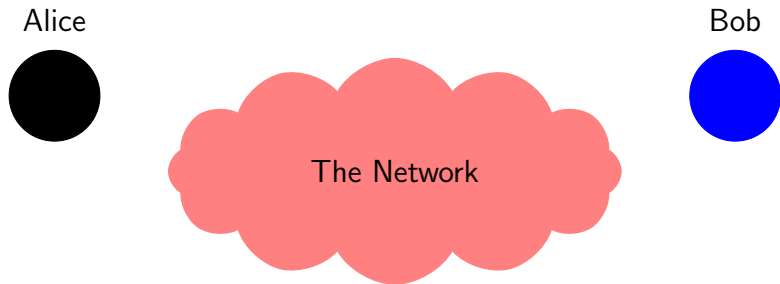
CPSA adversarial strand actions

- M. Text Message: $\langle +t \rangle$ where $t \in T$
- T. Tee: $\langle -g, +g, +g \rangle$ where $g \in G$
- C. Concatenation: $\langle -g, -h, +gh \rangle$
where $g, h, gh \in G$
- S. Separation into components:
 $\langle -gh, +g, +h \rangle$
- K. Key: $\langle +K \rangle$ where $K \in \mathbf{K}_p$
- E. Encryption $\langle -K, -h, +\{h\}_K \rangle$
- D. Decryption $\langle -K^{-1}, -\{h\}_K, +h \rangle$

1. $\langle +m \rangle$ emit a message from the network.
2. $\langle -m \rangle$ extract a message from the network.
3. $\langle m_1 m_2 \rangle$ concatenate two messages.
4. $\langle \{m\}_K \rangle$ encrypt a message m under key K .

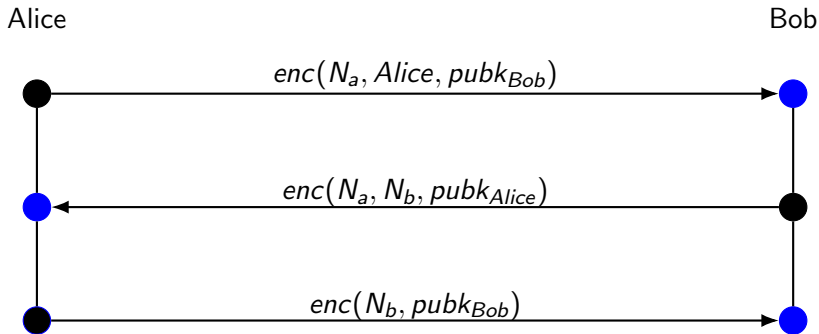
An adversarial model of the network

An adversarial model of the network



Goal: Mutually *authenticate* (verify identities) over an insecure network.

Needham Schroeder



Is this protocol secure against the Dolev-Yao Adversary?

Blue circle = Reception Black circle = Transmission

How can we spot attacks?

How can we spot attacks?

► *SECURITY GOALS*

How can we spot attacks?

- ▶ *SECURITY GOALS*
- ▶ Dotted lines

How can we spot attacks?

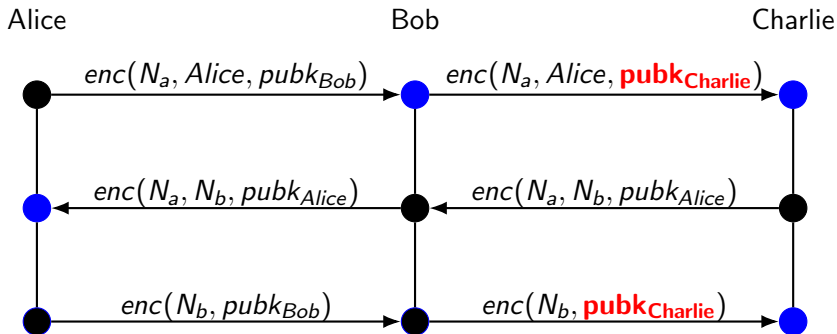
- ▶ *SECURITY GOALS*
- ▶ Dotted lines
- ▶ Listener Strands

How can we spot attacks?

- ▶ *SECURITY GOALS*
 - ▶ Dotted lines
 - ▶ Listener Strands
-
- ▶ What do we notice in the NS shapes from last time?

Needham Schroeder Attack

Needham Schroeder Attack



Remember: The Dolev-Yao Adversary can be a legitimate party on the network

Blue circle = Reception Black circle = Transmission

What do we give CPSA?

`protocol.scm`

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▶ `herald`

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`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`

What do we give CPSA?

protocol.scm

- ▶ herald
- ▶ defmacro
- ▶ defprotocol
 - ▶ defrole

What do we give CPSA?

`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`
- ▶ `defprotocol`
 - ▶ `defrole`
- ▶ `defskeleton`
- ▶ `defstrand`

What does CPSA give us?

`protocol.scm` →

What does CPSA give us?

`protocol.scm` →

▶ `protocol.txt`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`
- ▶ `protocol.xhtml`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`
- ▶ `protocol.xhtml`
- ▶ `protocol_shapes.xhtml`

Homework

Find attacks in CPSA

Find attacks in CPSA

Looking at shapes to find attacks takes some practice, so let us do some practice!

Two of the protocols are already done, you just have to run make so CPSA can make shapes, and then analyze them.

Two of the protocols need to be filled in, you will have to add part of a role, fill in a skeleton, or something else (documented in the comments), and then run CPSA to analyze the shapes.

Survey

Fill in details about the survey here!